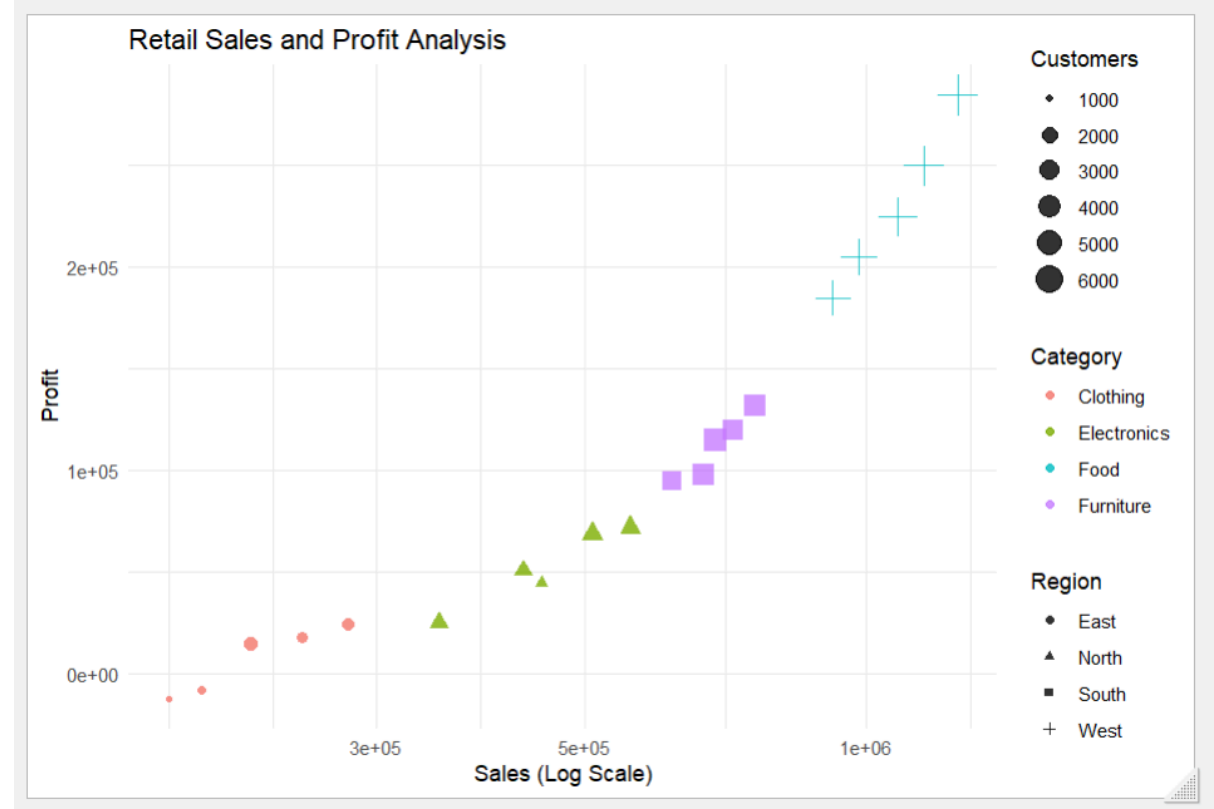
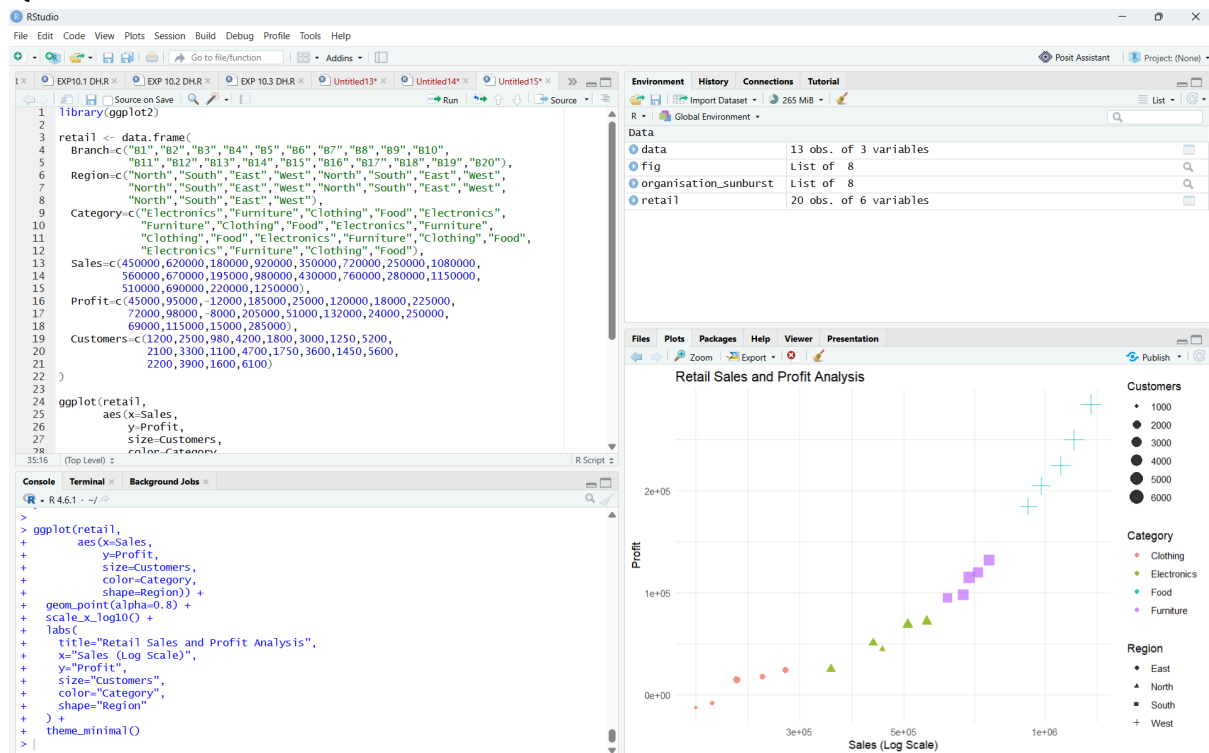


CO1_AT3_Visualization Design Exercise

Question 1

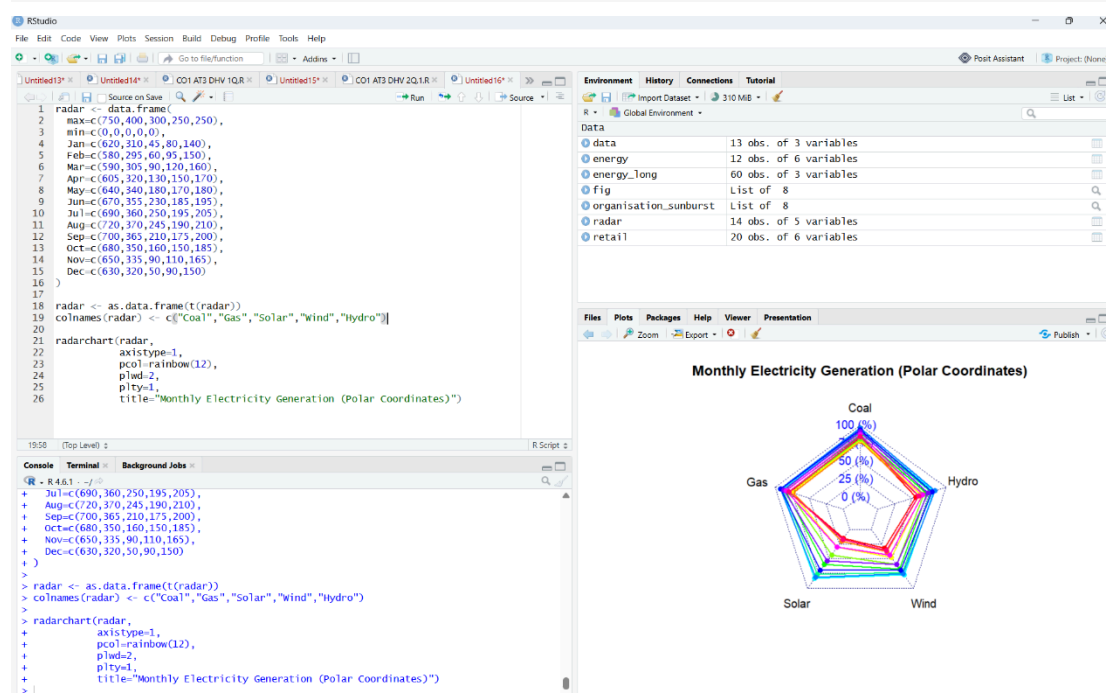
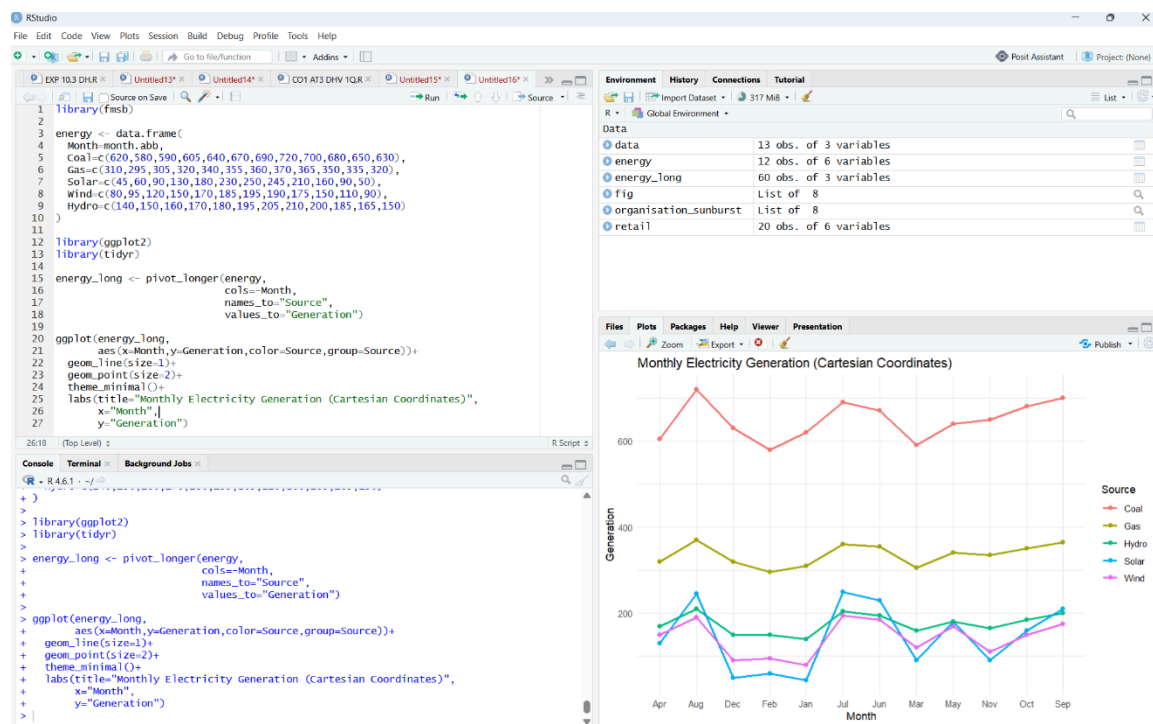


EXPLANATION:

1. **Bubble Chart** is chosen because it displays multiple variables in one visualization.

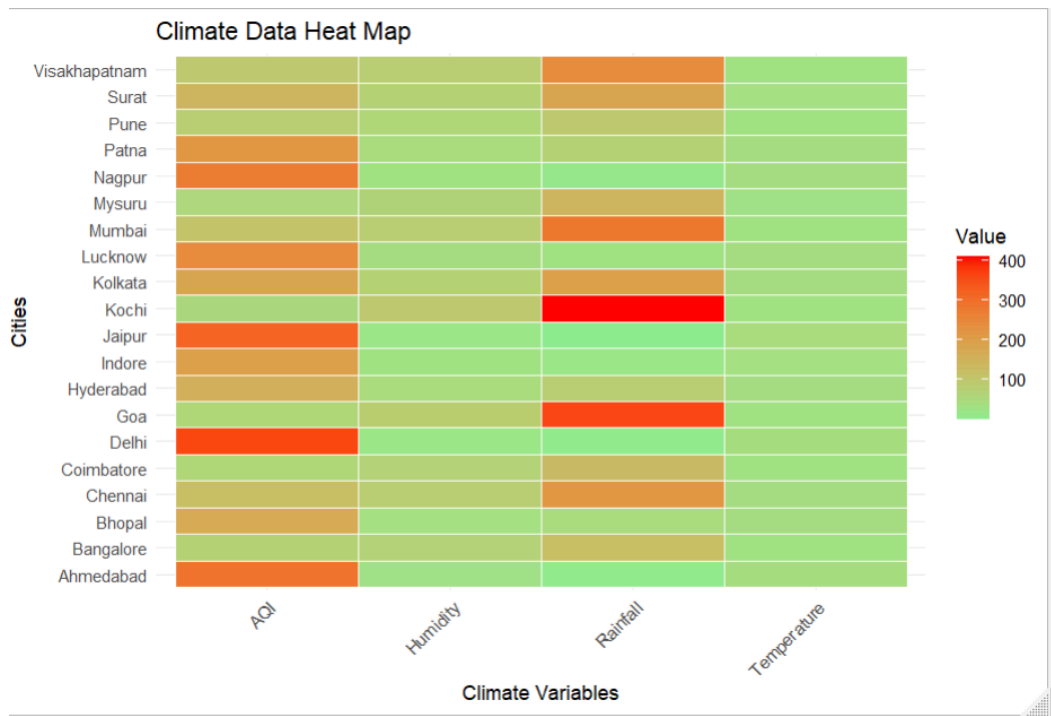
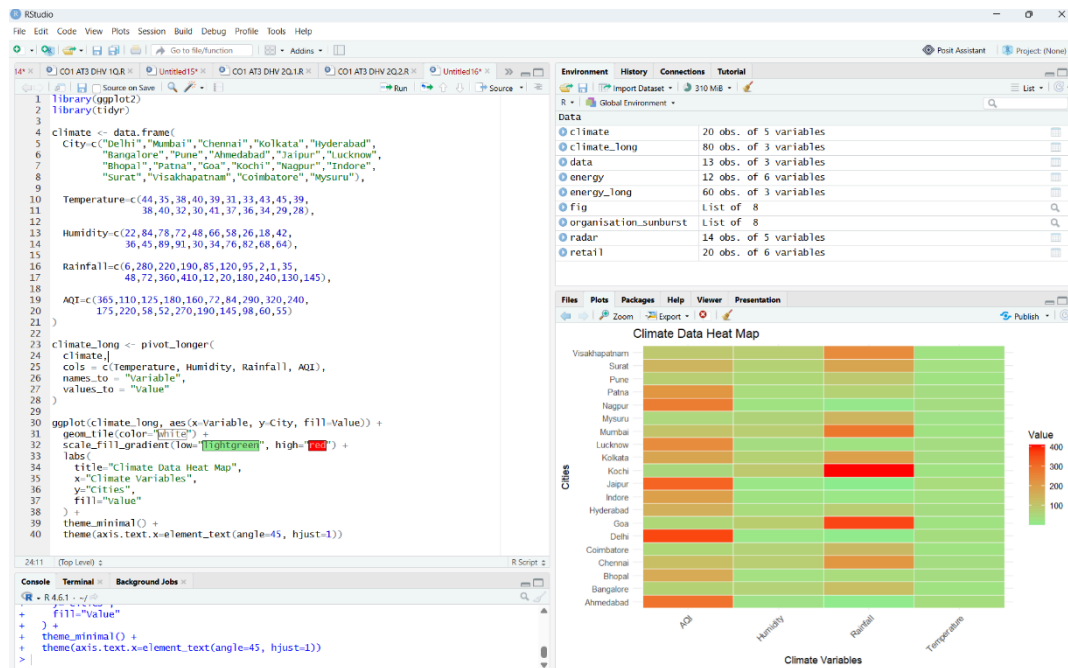
2. **X-axis (Sales)** shows branch sales.
3. **Y-axis (Profit)** compares branch profit.
4. **Bubble size** represents the number of customers.
5. **Bubble color** distinguishes product categories.
6. **Bubble shape** identifies different regions.
7. **Logarithmic scale** on Sales improves readability since Sales values are much larger than Profit values.
8. **Minimal theme** provides a clean and professional appearance.

QUESTION 2:



- **Cartesian Coordinates:** Best for comparing monthly trends and identifying increases or decreases.
- **Polar Coordinates:** Best for observing seasonal and cyclical patterns.
- **Conclusion:** Cartesian coordinates provide better analytical insight because values and trends are easier to compare accurately

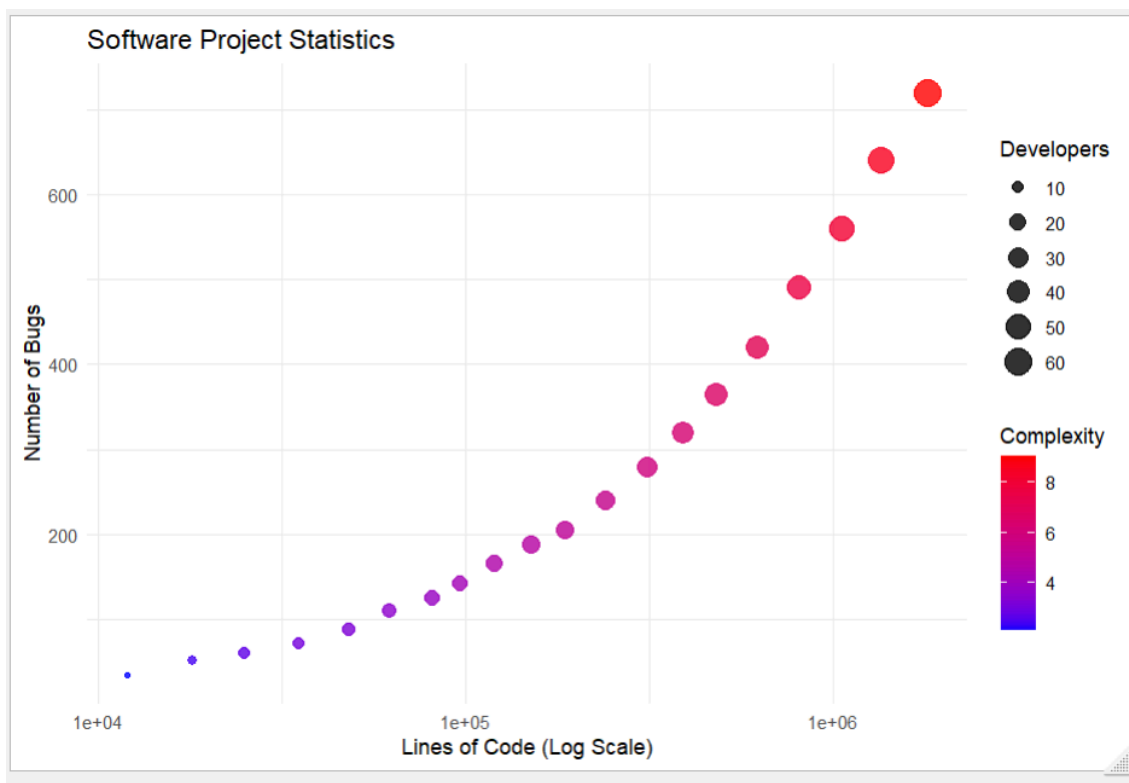
QUESTION 3:

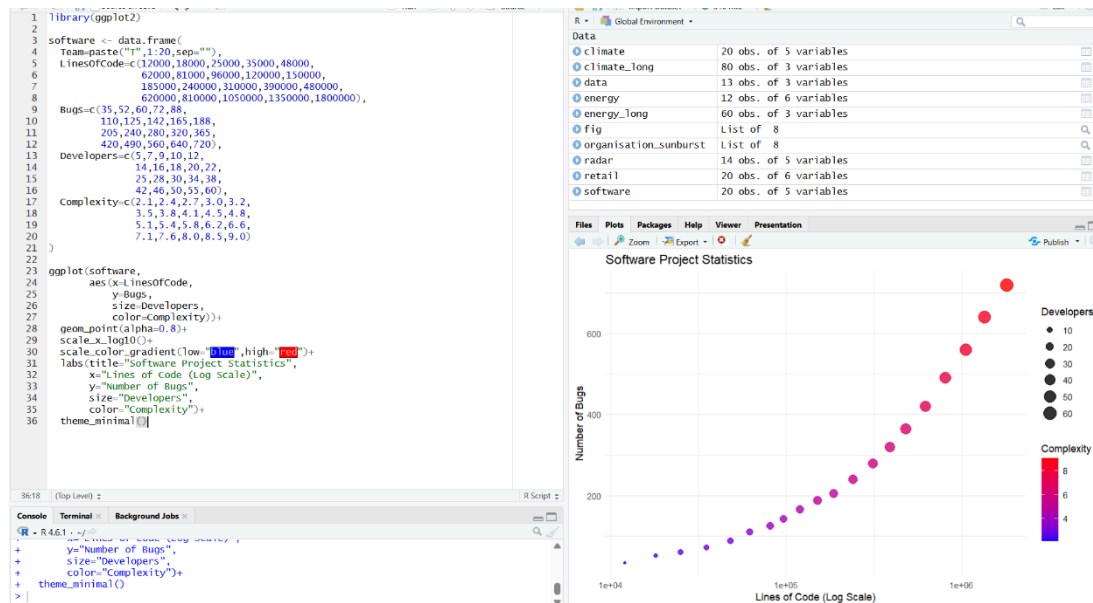


EXPLANATION:

- A Heat Map is selected because it effectively compares multiple climate variables across many cities.
- Cities are displayed on the Y-axis and climate variables on the X-axis.
- The colour intensity represents the magnitude of each value.
- A light yellow to red colour palette is used, where lighter colours indicate lower values and darker red indicates higher values.
- The minimal theme improves readability by reducing unnecessary visual elements.

QUESTION 4:

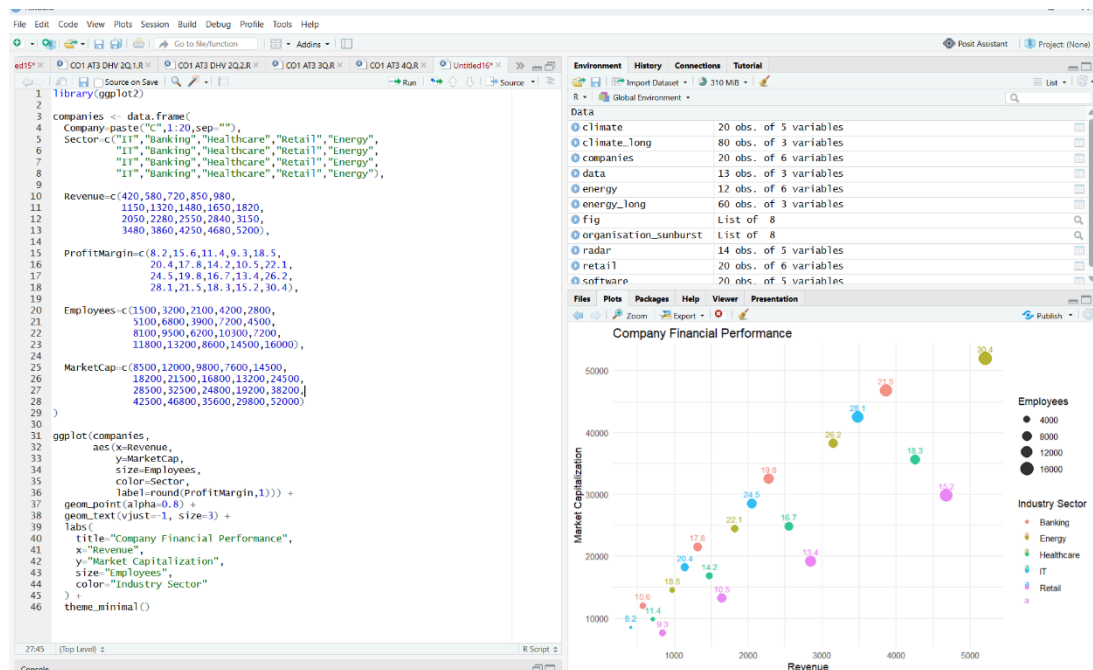


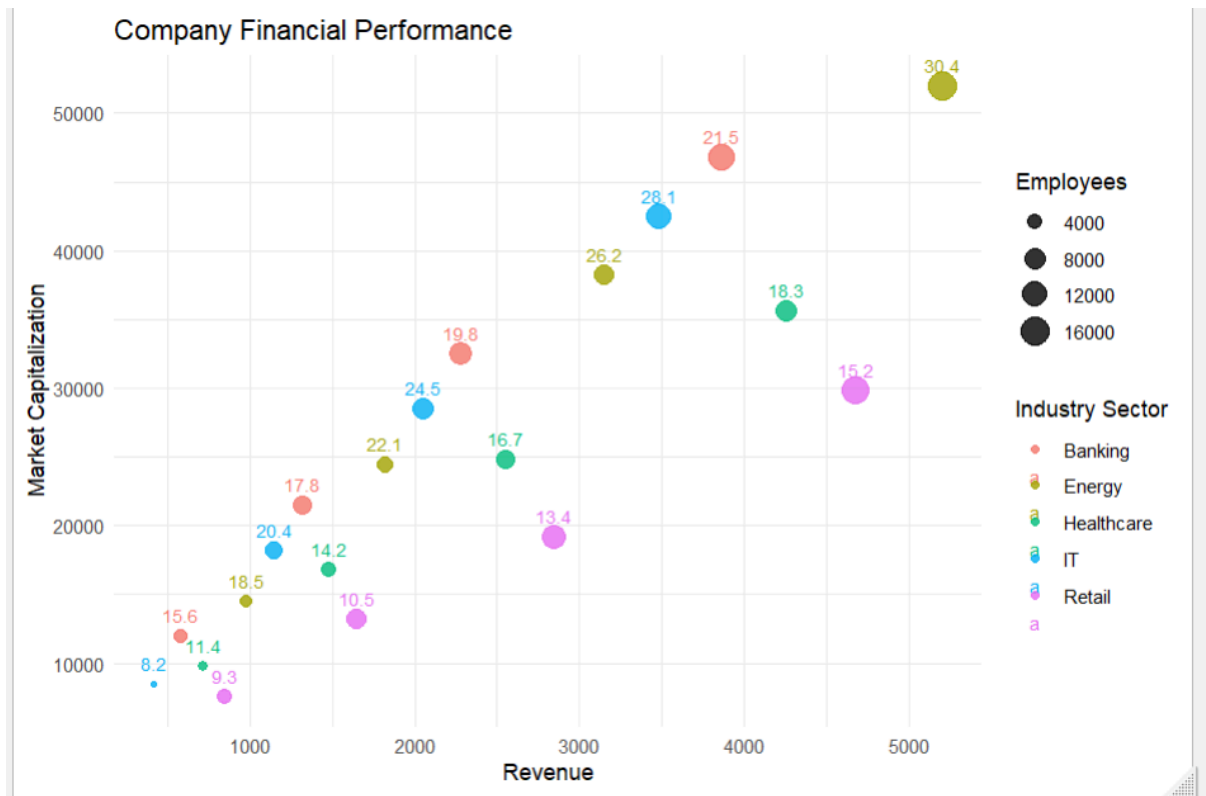


EXPLANATION:

- A Bubble Chart is chosen because it displays four variables in one visualization.
- X-axis: Lines of Code (logarithmic scale).
- Y-axis: Number of Bugs.
- Bubble Size: Number of Developers.
- Bubble Color: Project Complexity.
- A log10 transformation is applied because the Lines of Code values are highly skewed.

QUESTION 5:





Explanation:

- The chart allows simultaneous comparison of Revenue, Market Capitalization, Employees, Profit Margin, and Industry Sector.
- Larger bubbles indicate companies with more employees.
- Different colors help identify industry sectors quickly.
- Profit Margin labels allow direct comparison of company profitability.
- The visualization makes it easy to identify high-revenue, high-market-cap companies and compare their workforce and profitability.